

DBSE-DBD Practical Week 3

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Query 1: Create the book_metadata table

The screenshot shows the SQL Enterprise Manager interface with the following details:

- Navigator:** Schemas list includes bookflow_db, practical_week4, practical_week_1, practical_week_2, skill_week_1, skill_week_2, and sys.
- Query 1:**

```
1 CREATE DATABASE Practical_Week_3;
2
3 USE Practical_Week_3;
4
5 CREATE TABLE book_metadata (
6     book_id INT PRIMARY KEY,
7     title VARCHAR(255),
8     author VARCHAR(255),
9     published_year INT,
10    category VARCHAR(100),
11    format VARCHAR(50),
12    file_size VARCHAR(50),
13    height VARCHAR(20),
14    width VARCHAR(20),
15    weight VARCHAR(20),
16    member_id INT
17 );
```
- Output:**

#	Time	Action	Message	Duration / Fetch
1	14:58:04	CREATE DATABASE Practical_Week_3	1 row(s) affected	0.015 sec
2	14:58:04	USE Practical_Week_3	0 row(s) affected	0.000 sec
3	14:58:04	CREATE TABLE book_metadata (book_id INT PRIMARY KEY, title VARCHAR(255), author VARCHAR(255), published_year INT, category VARCHAR(100), format VARCHAR(50), file_size VARCHAR(50), height VARCHAR(20), width VARCHAR(20), weight VARCHAR(20), member_id INT)	0 row(s) affected	0.047 sec
- SQLAdditions:** Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Query 2: Create the book_reviews table

The screenshot shows the SQL Enterprise Manager interface with the following details:

- Navigator:** Schemas list includes bookflow_db, practical_week4, practical_week_1, practical_week_2, skill_week_1, skill_week_2, and sys.
- Query 1:**

```
1 CREATE TABLE book_reviews (
2     review_id INT AUTO_INCREMENT PRIMARY KEY,
3     book_id INT,
4     member_id INT,
5     rating INT,
6     comments TEXT,
7     FOREIGN KEY (book_id) REFERENCES book_metadata(book_id)
8 );
9
```
- Output:**

#	Time	Action	Message	Duration / Fetch
1	14:58:20	CREATE TABLE book_reviews (review_id INT AUTO_INCREMENT PRIMARY KEY, book_id INT, member_id INT, rating INT, comments TEXT, FOREIGN KEY (book_id) REFERENCES book_metadata(book_id))	0 row(s) affected	0.125 sec
- SQLAdditions:** Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Query 3: Insert book metadata

The screenshot shows the SQL Developer interface with the following components:

- Navigator:** Schemas list includes bookflow_db, practical_week4, practical_week1, practical_week2, skill_week1, skill_week2, and sys.
- Query Editor:** Contains the following SQL query:

```
1 INSERT INTO book_metadata (book_id, title, author, published_year, category, format, file_size, height, width, weight, me
2 (101, 'The Martian', 'Andy Weir', 2021, 'Science Fiction', 'Digital', '2.5', NULL, NULL, NULL, 101),
3 (102, 'Dune', 'Frank Herbert', 2022, 'Science Fiction', 'Digital', NULL, NULL, NULL, NULL, 102),
4 (103, 'Clean Code', 'Robert C. Martin', 2019, 'Programming', 'Physical', NULL, '22', '15', '1.5', 103),
5 (104, 'Artificial Intelligence Basics', 'John Smith', 2023, 'Artificial Intelligence', 'Digital', '4.2', NULL, NULL, NULL,
6 (105, 'Data Structures', 'Mark Allen', 2024, 'Computer Science', 'Digital', 'PDF', NULL, NULL, NULL, 105);
```
- Output:** Shows the execution results:

#	Time	Action	Message	Duration / Fetch
1	14:58:39	INSERT INTO book_metadata (book_id, title, author, published_year, category, format, file_size, height, width, weight, me	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.047 sec
- SQLAdditions:** Displays a message: "Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help."

Query 4: Insert book reviews

The screenshot shows the SQL Developer interface with the following components:

- Navigator:** Schemas list includes bookflow_db, practical_week4, practical_week1, practical_week2, skill_week1, skill_week2, and sys.
- Query Editor:** Contains the following SQL query:

```
1 INSERT INTO book_reviews (book_id, member_id, rating, comments) VALUES
2 (101, 101, 5, 'Excellent science fiction story'),
3 (101, 102, 4, 'Very interesting and informative'),
4 (101, 103, 5, 'Excellent space adventure'),
5 (102, 101, 5, 'Amazing world building'),
6 (102, 104, 4, 'Classic science fiction novel'),
7 (102, 105, 5, 'Excellent characters and story'),
8 (103, 101, 5, 'Excellent programming reference'),
9 (103, 106, 4, 'Very useful coding practices'),
10 (104, 102, 4, 'Good introduction to artificial intelligence'),
11 (104, 105, 5, 'Excellent machine learning explanation'),
12 (105, 101, 5, 'Excellent data structures book'),
13 (105, 106, 4, 'Very helpful for beginners'),
14 (105, 107, 4, 'Good examples and strict explanations'),
15
```
- Output:** Shows the execution results:

#	Time	Action	Message	Duration / Fetch
1	14:58:57	INSERT INTO book_reviews (book_id, member_id, rating, comments) VALUES	13 row(s) affected Records: 13 Duplicates: 0 Warnings: 0	0.046 sec
- SQLAdditions:** Displays a message: "Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help."

Query 5: Select all records from book_metadata

2520090085_Vakkalanka_Sai... x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- bookflow_db
- practical_week4
- practical_week_1
- practical_week_2
- skill_week_1
- skill_week_2
- sys

Query 1

1 • SELECT * FROM book_metadata;

Limit to 1000 rows

Result Grid

book_id	title	author	published_year	category	format	file_size	height	width	weight	member_id
101	The Martian	Andy Weir	2021	Science Fiction	Digital	2.5 MB	100%	100%	100%	101
102	Dune	Frank Herbert	2022	Science Fiction	Digital	100%	100%	100%	100%	102
103	Clean Code	Robert C. Martin	2019	Programming	Physical	22	15	1.5	103	103
104	Artificial Intelligence Basics	John Smith	2023	Artificial Intelligence	Digital	4.2 MB	100%	100%	100%	104
105	Data Structures	Mark Allen	2024	Computer Science	Digital	PDF	100%	100%	100%	105

ok_metadata 1 x

Apply Revert Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	14:59:26	SELECT * FROM book_metadata LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec

No object selected

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Query 6: Select all records from book_reviews

2520090085_Vakkalanka_Sai... x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- bookflow_db
- practical_week4
- practical_week_1
- practical_week_2
- skill_week_1
- skill_week_2
- sys

Query 1

1 • SELECT * FROM book_reviews;

Execute the selected portion of the script or everything, if there is no selection

Limit to 1000 rows

Result Grid

review_id	book_id	member_id	rating	comments
1	101	101	5	Excellent science fiction story
2	101	102	4	Very interesting and informative
3	101	103	5	Excellent space adventure
4	102	101	5	Amazing world building
5	102	104	4	Classic science fiction novel
6	102	105	5	Excellent characters and story
7	103	101	5	Excellent programming reference
8	103	106	4	Very useful coding practices

book_reviews 2 x

Apply Revert Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	14:59:45	SELECT * FROM book_reviews LIMIT 0, 1000	13 row(s) returned	0.000 sec / 0.000 sec

No object selected

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Query 7: Books with rating greater than 4

Query 1

```
1 SELECT DISTINCT b.book_id, b.title, b.author
2 FROM book_metadata b
3 JOIN book_reviews r
4 ON b.book_id = r.book_id
5 WHERE r.rating > 4;
```

Result Grid

book_id	title	author
101	The Martian	Andy Weir
102	Dune	Frank Herbert
103	Clean Code	Robert C. Martin
104	Artificial Intelligence Basics	John Smith
105	Data Structures	Mark Allen

Result 5

Output

#	Time	Action	Message	Duration / Fetch
1	15:00:23	SELECT DISTINCT b.book_id, b.title, b.author FROM book_metadata b JOIN book_reviews r ON b.book_id = r...	5 row(s) returned	0.031 sec / 0.000 sec

Query 8: Books with rating IN (4, 5)

Query 1

```
1 SELECT DISTINCT b.book_id, b.title, b.author
2 FROM book_metadata b
3 JOIN book_reviews r
4 ON b.book_id = r.book_id
5 WHERE r.rating IN (4, 5);
```

Result Grid

book_id	title	author
101	The Martian	Andy Weir
102	Dune	Frank Herbert
103	Clean Code	Robert C. Martin
104	Artificial Intelligence Basics	John Smith
105	Data Structures	Mark Allen

Result 6

Output

#	Time	Action	Message	Duration / Fetch
1	15:01:35	SELECT DISTINCT b.book_id, b.title, b.author FROM book_metadata b JOIN book_reviews r ON b.book_id = r...	5 row(s) returned	0.016 sec / 0.000 sec

Query 9: Books with Digital format

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File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- bookflow_db
- practical_week4
- practical_week_1
- practical_week_2
- skill_week_1
- skill_week_2
- sys

Query 1

```
1 SELECT *
2 FROM book_metadata
3 WHERE format LIKE 'Digital%';
```

Result Grid

book_id	title	author	published_year	category	format	file_size	height	width	weight	member_id
101	The Martian	Andy Weir	2021	Science Fiction	Digital	2.5 MB	1028	6028	0.028	101
102	Dune	Frank Herbert	2022	Science Fiction	Digital	4.2 MB	1028	6028	0.028	102
104	Artificial Intelligence Basics	John Smith	2023	Artificial Intelligence	Digital	4.2 MB	1028	6028	0.028	104
105	Data Structures	Mark Allen	2024	Computer Science	Digital	PDF	1028	6028	0.028	105

book_metadata 7 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	15:01:53	SELECT * FROM book_metadata WHERE format LIKE 'Digital%'; LIMIT 0, 1000	4 row(s) returned	0.016 sec / 0.000 sec

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Query 10: Average rating and total reviews

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File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- bookflow_db
- practical_week4
- practical_week_1
- practical_week_2
- skill_week_1
- skill_week_2
- sys

Query 1

```
1 SELECT
2     b.title,
3     ROUND(AVG(r.rating), 2) AS average_rating,
4     COUNT(r.review_id) AS total_reviews
5 FROM book_metadata b
6 JOIN book_reviews r
7 ON b.book_id = r.book_id
8 WHERE b.published_year > 2020
9 GROUP BY b.book_id, b.title
10 ORDER BY average_rating DESC;
```

Result Grid

title	average_rating	total_reviews
The Martian	4.67	3
Dune	4.67	3
Artificial Intelligence Basics	4.50	2
Data Structures	4.33	3

Result 8 x

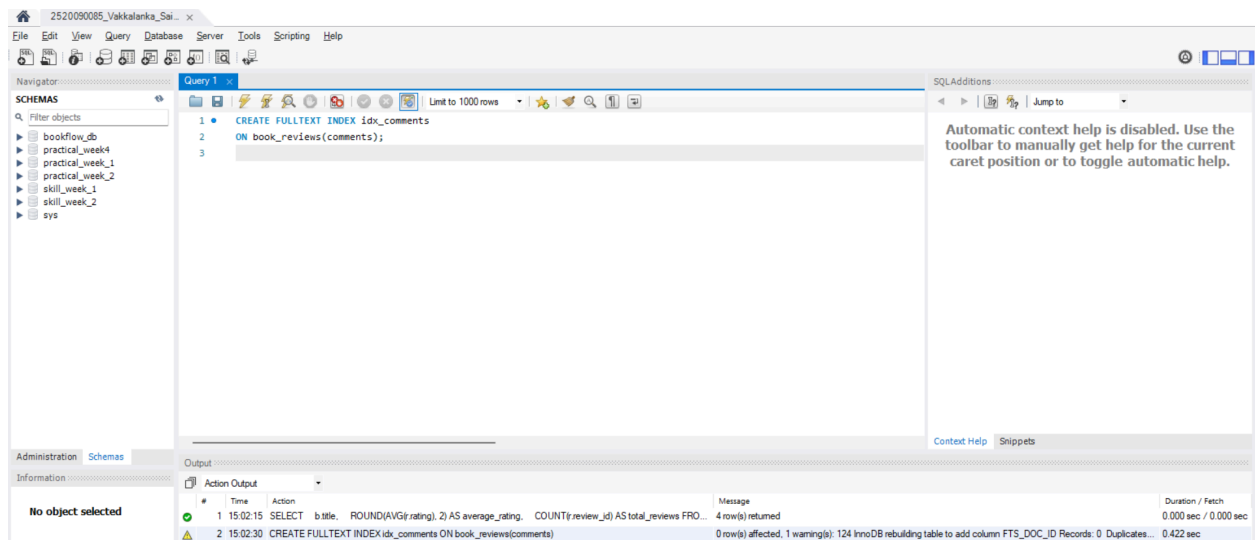
Output

Action Output

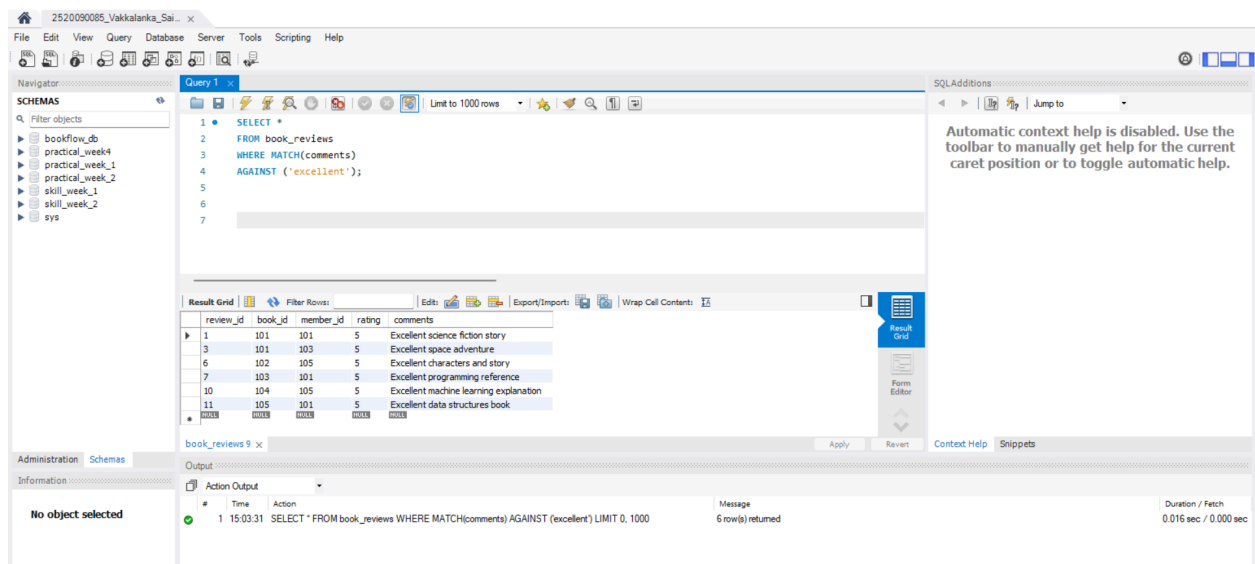
#	Time	Action	Message	Duration / Fetch
1	15:02:15	SELECT b.title, ROUND(AVG(r.rating), 2) AS average_rating, COUNT(r.review_id) AS total_reviews FROM ...	4 row(s) returned	0.000 sec / 0.000 sec

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Query 11: Create FULLTEXT index on comments



Query 12: FULLTEXT search for 'excellent'



Query 13: FULLTEXT search for 'science fiction'

2520090085_Vakkalanka_Sai... x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- bookflow_db
- practical_week4
- practical_week_1
- practical_week_2
- skill_week_1
- skill_week_2
- sys

Query 1

```
1 SELECT *
2 FROM book_reviews
3 WHERE MATCH(comments)
4 AGAINST ('science fiction');
```

Limit to 1000 rows

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Result Grid

review_id	book_id	member_id	rating	comments
1	101	101	5	Excellent science fiction story
5	102	104	4	Classic science fiction novel

book_reviews 10 x

Apply Revert Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	15:03:55	SELECT * FROM book_reviews WHERE MATCH(comments) AGAINST ('science fiction') LIMIT 0, 1000	2 row(s) returned	0.016 sec / 0.000 sec

No object selected

Query 14: EXPLAIN FULLTEXT search

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File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- bookflow_db
- practical_week4
- practical_week_1
- practical_week_2
- skill_week_1
- skill_week_2
- sys

Query 1

```
1 EXPLAIN
2 SELECT *
3 FROM book_reviews
4 WHERE MATCH(comments)
5 AGAINST ('excellent');
```

Limit to 1000 rows

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Result Grid

id	select_type	table	partitions	type	possible_keys	key	key_len	ref	rows	filtered	Extra
1	SIMPLE	book_reviews		fulltext	idx_comments	idx_comments	0	const	1	100.00	Using where; Pt_hints: sorted

Result 11 x

Read Only

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	15:04:09	EXPLAIN SELECT * FROM book_reviews WHERE MATCH(comments) AGAINST ('excellent')	1 row(s) returned	0.031 sec / 0.000 sec

No object selected